Computing Nash equilibria is computationally hard. Daskalakis et al. (2009) showed that finding a Nash equilibrium in two-player normal-form games is PPAD-complete, meaning no polynomial-time algorithm is known and none is likely to exist. Extensive-form games with imperfect information inherit this hardness and introduce additional challenges from sequential decision-making and hidden information.[1] Counterfactual regret minimization (CFR) achieves $O(1/\sqrt{T})$ convergence to ε-Nash equilibrium in two-player zero-sum games (Zinkevich et al., 2008). After T iterations, exploitability decreases as $1/\sqrt{T}$, so reaching exploitability 0.01 requires approximately 10,000 iterations.[2] This section applies CFR to the durable goods monopoly problem, benchmarking against the theoretical results of Gul et al. (1986) and Ausubel and Deneckere (1989).

0.1 Counterfactual Regret Minimization

An extensive-form game consists of a game tree with information sets $\mathcal{I}_i$ partitioning player i's decision nodes. An information set groups nodes where the player cannot distinguish between them due to hidden opponent actions or private information. A behavioral strategy $\sigma_i : \mathcal{I}_i \to \Delta(\mathcal{A})$ assigns action probabilities at each information set.

The key concept in CFR is the counterfactual value, which measures how much better player i would have done by taking action a at information set I, assuming she played to reach I but her opponent played according to σ? Formally:

$$v_i^\sigma(I, a) = \sum_{h \in I} \pi_{-i}^\sigma(h) \sum_{z \sqsupseteq ha} \pi^\sigma(ha, z) u_i(z) \tag{1}$$

where $\pi_{-i}^\sigma(h)$ is the probability opponents reach h, and $\pi^\sigma(ha, z)$ is the probability of reaching terminal z from ha. The counterfactual formulation weights by opponent reach $\pi_{-i}^\sigma(h)$ rather than joint reach $\pi^\sigma(h)$. This ensures non-zero updates even at rarely-visited information sets, avoiding the vanishing gradient problem that affects direct policy gradient methods in extensive-form games. Cumulative regret for action a is $R^T(I, a) =$

[1] PPAD (Polynomial Parity Arguments on Directed graphs) is a complexity class for problems where existence is guaranteed by a parity argument. PPAD-completeness means the problem is as hard as any problem in this class. For comparison, checking whether a strategy profile is a Nash equilibrium is easy (polynomial time); finding one is PPAD-complete.

[2] An ε-Nash equilibrium is a strategy profile where no player can improve her expected payoff by more than ε through unilateral deviation. As $\varepsilon \to 0$, this converges to an exact Nash equilibrium.

$\sum_{t=1}^{T}[v^{\sigma^t}(I,a) - v^{\sigma^t}(I)]$. CFR updates via regret matching:[3]

$$\sigma^{T+1}(I,a) = \frac{[R^T(I,a)]^+}{\sum_{a'}[R^T(I,a')]^+}, \quad [x]^+ = \max(x,0). \tag{2}$$

The average strategy $\bar{\sigma}^T$ converges to ε-Nash with $\varepsilon = O(|\mathcal{I}|\sqrt{|\mathcal{A}|}\Delta/\sqrt{T})$, where Δ is the range of payoffs (Zinkevich et al., 2008).

CFR+ (Tammelin, 2014) replaces standard regret matching with Regret Matching+, which truncates cumulative regrets to zero after each update rather than only at action selection. The update becomes $R^{T+1}(I,a) = \max(R^T(I,a) + r^{T+1}(I,a), 0)$, where $r^{T+1}(I,a) = v^{\sigma^{T+1}}(I,a) - v^{\sigma^{T+1}}(I)$ is the instantaneous counterfactual regret. CFR+ also weights iteration t by t when computing the average strategy, giving later iterations, which reflect strategies closer to equilibrium, more influence. While vanilla CFR converges at $O(1/\sqrt{T})$, CFR+ empirically converges at $O(1/T)$, roughly squaring the number of useful digits of exploitability per unit of compute.[4] CFR+ enabled Bowling et al. (2015) to essentially solve heads-up limit Texas hold'em, a game with 3.16×10^{17} states, the first non-trivial imperfect-information game played competitively by humans to be essentially solved. Their program Cepheus achieved exploitability below 1 mbb/g (milli-big-blind per game), small enough that a human lifetime of play would not suffice to distinguish it from a Nash equilibrium strategy.

0.2 Neural Extensions

Tabular CFR stores regrets at every information set, infeasible when $|\mathcal{I}| > 10^{14}$. Two neural approaches scale to large games.

0.2.1 Deep CFR

Brown et al. (2019) approximate cumulative regrets with a neural network $V_\theta : \mathcal{I} \times \mathcal{A} \to \mathbb{R}$ trained on sampled (information set, iteration, regret) tuples:

$$L_V(\theta) = \mathbb{E}_{(I,t',r)\sim\mathcal{M}}\left[t' \cdot (V_\theta(I,a) - r)^2\right]. \tag{3}$$

[3]Regret matching is an action selection rule where the probability of choosing action a is proportional to the cumulative regret for not having chosen a in the past (truncated at zero). Actions with high regret receive higher probability; actions with negative regret (having performed worse than average) receive zero probability.

[4]The $O(1/T)$ rate for CFR+ is empirically observed but not yet proven in full generality. Tammelin (2014) conjectured this rate based on extensive experiments across poker variants.

Weighting by iteration index t' gives more recent regret estimates higher importance, since later iterations reflect strategies closer to equilibrium. A separate network Π_ϕ approximates the average strategy, trained via weighted MSE on strategy samples with the same iteration weighting. The current strategy derives from regret matching, with $\sigma(I, a) \propto [V_\theta(I, a)]^+$.

0.2.2 Neural Fictitious Self-Play

NFSP (Heinrich and Silver, 2016) combines fictitious play[5] (Brown, 1951) with deep Q-learning.[6] Each player maintains two networks: a best-response network Q_θ trained via DQN, and an average strategy network $\bar{\pi}_\phi$ trained via supervised learning on the best-response actions:

$$L_{\mathrm{RL}}(\theta) = \mathbb{E}\left[\left(r + \gamma \max_{a'} Q_{\theta^-}(I', a') - Q_\theta(I, a)\right)^2\right], \quad L_{\mathrm{SL}}(\phi) = \mathbb{E}\left[-\log \bar{\pi}_\phi(a|I)\right]. \tag{4}$$

During play, agents follow $\bar{\pi}_\phi$ with probability $1 - \eta$ and Q_θ with probability η.

0.2.3 Poker Results

These methods achieved superhuman performance in poker. Deep CFR attained exploitability of 37 mbb/g (milli-big-blinds per game)[7] in heads-up flop hold'em (Brown et al., 2019). Libratus defeated top human professionals in heads-up no-limit hold'em using nested subgame solving with CFR (Brown and Sandholm, 2018). Pluribus extended to six-player no-limit hold'em (Brown and Sandholm, 2019). The game tree of heads-up no-limit hold'em contains $\sim 10^{161}$ states; these results demonstrate that CFR-based methods scale to economically relevant game sizes.

[5] Fictitious play (Brown, 1951) is a classical learning rule where each player best-responds to the empirical distribution of opponents' past actions. Under certain conditions (e.g., two-player zero-sum games), the time-average strategies converge to Nash equilibrium.

[6] Deep Q-Network (DQN) (Mnih et al., 2015) approximates the Q-function with a neural network, using experience replay and a target network to stabilize training. The target network θ^- in the loss function is a delayed copy of the main network, updated periodically.

[7] Exploitability measures how far a strategy is from Nash equilibrium, defined as the maximum expected gain an adversary could achieve by best-responding. Zero exploitability means the strategy is unexploitable (Nash). In poker, exploitability is measured in milli-big-blinds per game (mbb/g).

0.3 The Coase Conjecture

The durable goods monopoly problem has two distinct cases that yield qualitatively different equilibria. In the "gap case," the seller's cost is strictly below all buyer valuations, so trade is guaranteed. In the "no-gap case," cost equals or exceeds the lowest buyer valuation, so some buyers may never trade. The gap case admits a unique stationary equilibrium; the no-gap case admits multiple equilibria with more complex dynamics. I study the gap case, which has a clean theoretical benchmark.

Coase (1972) conjectured that a durable goods monopolist[8] loses market power when buyers are patient. Unable to commit to future prices, the seller competes with her future self, eroding rents. Stokey (1981) showed that a monopolist facing rational consumers cannot price discriminate intertemporally; Bulow (1982) proved that in the no-gap case, the monopolist prefers renting to selling. Gul et al. (1986) provided the definitive formalization. With a continuum of buyer types and continuous time, as the inter-offer interval $\Delta t \to 0$, price converges to marginal cost instantaneously and all trade occurs at $t = 0$.

0.3.1 Model

A seller with zero cost faces a buyer with private valuation $v \in \{v_L, v_H\}$, where $\Pr(v = v_H) = \pi$. In each period $t = 1, 2, \ldots$, the seller posts price p_t; the buyer accepts or rejects. Upon acceptance at t, the seller receives $\delta^{t-1} p_t$ and the buyer receives $\delta^{t-1}(v - p_t)$, where $\delta \in (0, 1)$ is the common discount factor. The game ends upon acceptance or after T periods.

This is the gap case, where seller cost (0) is strictly below the lowest buyer valuation (v_L), guaranteeing trade in equilibrium.

0.3.2 Theoretical Benchmark

The screening price $P^*(\delta)$ makes the high-type buyer indifferent between accepting now and waiting:

$$v_H - P^* = \delta(v_H - v_L) \implies P^*(\delta) = v_H - \delta(v_H - v_L). \tag{5}$$

[8] A durable good provides utility over multiple periods (e.g., a car, appliance, or software license) rather than being consumed immediately. Unlike non-durable goods, durable goods create intertemporal competition, as the seller at time t competes with her own future self at $t + 1$.

The seller's optimal strategy depends on π. Let $\Pi_{\text{screen}} = \pi P^* + (1-\pi)\delta v_L$ and $\Pi_{\text{pool}} = v_L$. The seller screens if $\Pi_{\text{screen}} > \Pi_{\text{pool}}$, yielding threshold

$$\pi^* = \frac{v_L(1-\delta)}{P^* - \delta v_L} = \frac{1}{2} \quad \text{when } v_H = 2v_L. \tag{6}$$

For $\pi < \pi^*$, the seller pools (offer v_L immediately). For $\pi > \pi^*$, the seller screens (offer P^*, then v_L if rejected).[9]

The Coase conjecture manifests as $\delta \to 1$, where the screening price $P^*(\delta) \to v_L$, so the seller cannot extract surplus from high types.

0.3.3 Computational Results

I model the bargaining game as an extensive-form game and apply CFR. Parameters: $v_L = 100$, $v_H = 200$, $T = 2$ periods. The seller's information set is the rejection history; the buyer's information set includes her private type.

Table 1: CFR Equilibrium vs. Theory: π-sweep at $\delta = 0.5$

π	P^*	P(Screen)	Theory	NashConv	Eq. Type	Status
0.10	150	0.000	0.0	5.0258	Pooling	✓
0.15	150	0.000	0.0	7.5249	Pooling	✓
0.20	150	0.000	0.0	10.0239	Pooling	✓
0.25	150	0.000	0.0	12.5229	Pooling	✓
0.30	150	0.000	0.0	15.0223	Pooling	✓
0.35	150	0.000	0.0	17.5216	Pooling	✓
0.40	150	0.000	0.0	20.0213	Pooling	✓
0.45	150	0.000	0.0	22.5213	Pooling	✓
0.50	150	0.001	0.5	25.0221	Indifferent	✓
0.55	150	0.002	1.0	27.5256	Screening	✓
0.60	150	0.506	1.0	30.0066	Screening	✓
0.65	150	0.999	1.0	26.2553	Screening	✓
0.70	150	1.000	1.0	22.5027	Screening	✓
0.75	150	1.000	1.0	18.7531	Screening	✓
0.80	150	1.000	1.0	15.0067	Screening	✓
0.85	150	1.000	1.0	11.2576	Screening	✓
0.90	150	1.000	1.0	7.5089	Screening	✓

Notes: P(Screen) is the probability the seller offers $P^* = 150$ in period 1. Theory column gives the predicted probability (0 for pooling, 1 for screening). NashConv is the sum of exploitabilities. 5,000 CFR iterations per configuration.

Table 1 reports results from varying π at fixed $\delta = 0.5$. The algorithm recovers the

[9] In a screening equilibrium, the seller uses price to separate buyer types. High-value buyers accept immediately at a high price, while low-value buyers reject and receive a lower offer later. In a pooling equilibrium, the seller offers a single price that all types accept, sacrificing price discrimination for immediate sale.

phase transition at $\pi^* = 0.5$. For $\pi < 0.5$, the seller pools with probability approaching 1; for $\pi > 0.5$, the seller screens with probability approaching 1. The transition is sharp.

A δ-sweep at $\pi = 0.7$ confirms the screening price formula: $P^*(\delta) = 200 - 100\delta$ with zero error across $\delta \in [0.1, 0.9]$. As δ increases, screening becomes unprofitable (Coase conjecture); at $\delta = 0.75$, the seller switches to pooling.

0.3.4 Validation

Four stress tests verify the results.[10] The 3-period result merits explanation. At $P_1 = 136$, the high buyer is exactly indifferent: surplus $= v_H - 136 = 64 = \delta(v_H - 120)$. If the buyer randomizes, the seller prefers $P_1 = 120$ (revenue 108 vs. 86.4 if buyer rejects 136). CFR found the equilibrium robust to tie-breaking.

0.4 Discussion

CFR computed equilibria matching the theoretical predictions of Gul et al. (1986) without any knowledge of the economic structure. The algorithm discovered that (1) the seller screens when π is high, pools when π is low, with threshold at $\pi^* = 0.5$; (2) the screening price satisfies the high-buyer indifference condition; (3) as $\delta \to 1$, screening collapses (Coase conjecture). These properties emerged from regret minimization alone.

[10] (1) Awkward primes $(v_L, v_H) = (37, 83)$: converged to $P^* = 55$ (theory: 55.4). (2) Information leak: single seller information set at root. (3) Grid shift with 150 removed: used 145 (99.4%). (4) 3-period game: found $P_1 = 120$ vs. theory 136.

References

Lawrence M. Ausubel and Raymond J. Deneckere. Reputation in bargaining and durable goods monopoly. *Econometrica*, 57(3):511–531, 1989.

Michael Bowling, Neil Burch, Michael Johanson, and Oskari Tammelin. Heads-up limit hold'em poker is solved. *Science*, 347(6218):145–149, 2015.

George W. Brown. Iterative solution of games by fictitious play, 1951.

Noam Brown and Tuomas Sandholm. Superhuman ai for heads-up no-limit poker: Libratus beats top professionals. *Science*, 359(6374):418–424, 2018.

Noam Brown and Tuomas Sandholm. Superhuman ai for multiplayer poker. *Science*, 365 (6456):885–890, 2019.

Noam Brown, Adam Lerer, Sam Gross, and Tuomas Sandholm. Deep counterfactual regret minimization. In *Proceedings of the 36th International Conference on Machine Learning (ICML)*, volume 97, pages 793–802. PMLR, 2019.

Jeremy I. Bulow. Durable-goods monopolists. *Journal of Political Economy*, 90(2):314–332, 1982.

Ronald H. Coase. Durability and monopoly. *Journal of Law and Economics*, 15(1): 143–149, 1972.

Constantinos Daskalakis, Paul W. Goldberg, and Christos H. Papadimitriou. The complexity of computing a nash equilibrium. *SIAM Journal on Computing*, 39(1):195–259, 2009.

Faruk Gul, Hugo Sonnenschein, and Robert Wilson. Foundations of dynamic monopoly and the Coase conjecture. *Journal of Economic Theory*, 39(1):155–190, 1986.

Johannes Heinrich and David Silver. Deep reinforcement learning from self-play in imperfect-information games. *arXiv preprint arXiv:1603.01121*, 2016.

Volodymyr Mnih, Koray Kavukcuoglu, David Silver, Andrei A. Rusu, Joel Veness, Marc G. Bellemare, Alex Graves, Martin Riedmiller, Andreas K. Fidjeland, Georg Ostrovski, Stig Petersen, Charles Beattie, Amir Sadik, Ioannis Antonoglou, Helen King,

Dharshan Kumaran, Daan Wierstra, Shane Legg, and Demis Hassabis. Human-level control through deep reinforcement learning. *Nature*, 518:529–533, 2015.

Nancy L. Stokey. Rational expectations and durable goods pricing. *Bell Journal of Economics*, 12(1):112–128, 1981.

Oskari Tammelin. Solving large imperfect information games using cfr+. *arXiv preprint arXiv:1407.5042*, 2014.

Martin Zinkevich, Michael Johanson, Michael Bowling, and Carmelo Piccione. Regret minimization in games with incomplete information. In *Advances in Neural Information Processing Systems*, volume 20, 2008.